Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CE733]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

PAVEMENT DESIGN

Time: 3 Hours **Maximum Marks:70**

Instructions: i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.

ii. Missing data may be suitably assumed.

1. a. Outline the functions of pavement components. **06**

b. Differentiate between flexible and rigid pavement. **08**

2. a. Explain the plate bearing test procedure. How corrections for 'K' value may be made for a different plate size? **11**

b. What are the basic assumptions involved in pavement layered elastic model? **03**

3. a. Design a flexible pavement with the following data. **07**

Subgrade CBR=4%,

Moorum Subbase CBR=20%,

W.B.M base CBR=90%,

Expected traffic=600 vehicles per day. [Refer Figure 3.a]

b. Design the flexible pavement section by triaxial test method using the following data: Wheel load =4100Kg Radius of contact area = 15cm Traffic coefficient, X=1.5: Rainfall coefficient, Y=0.9: Design deflection, Δ=0.25cm: E- value of subgrade soil, Es=100Kg/cm^2 E- value of base course material, E_b = 400Kg/cm2: E- value of 7.5cm thick bituminous concrete surface = 1000Kg/cm^2 **07**

OR

4. Design a flexible pavement for a subgrade CBR value of 3.5. The expected number of repetitions of the standard axle load is 5 million. The reliability needed is 95%. The standard deviation is 0.35. The loss in serviceability expected is 1.9. The layer coefficient for asphalt surface course is 0.14. The stone aggregate base course has a CBR of 30 (a2=0.14) and of sub-base 100 (a3=0.11). The drainage condition of the base is good and that of the sub-base is excellent. The pavement is subjected to moisture condition approaching saturation to an extent of 20% of the time. Layer coefficient a1 for asphalt concrete surface course used in AASHTO is 0.44. Assume D2=6 inches. [Refer Figure 4.a and Table 4.a] **14**

5. For a cement concrete pavement, find the magnitude of wheel load stress at i)interior, ii)edge and iii)corner regions using Westergaard's stress equations for the following data and given charts. wheel load, P=5100Kg; modulus of elasticity of cement concrete, Ec=3x105kg/cm^2; pavement thickness, h=18cm; poisson's ratio of concrete, μ=0.15; modulus of subgrade reaction, K=6kg/cm3; radius of contact area, a=15cm. **14**

OR

6. a. Define warping stress? Indicate the Bradbury's equation used to determine warping **05**

stresses at the three critical locations.

b. Determine spacing between contraction joints for a 3.5m slab width having thickness of 200mm, coefficient of friction 1.5, for the following two conditions: **09**

i. Plain cement concrete, allowable Sc=0.8 Kg/cm^2

ii. Reinforced cement concrete, 1.0 cm diameter bars at 0.3m spacing

7. a. Give the reasons for the following defects with respect to rigid and flexible pavement **08**

i. Pot hole

ii. Alligator cracking

b. Briefly explain the various maintenance works of bituminous surfaces **06**

OR

8. a. What are the maintenance works required for cracks and joints in rigid pavements? **06**

b. Describe the method of design of overlays for flexible pavement as per IRC guidelines **08**

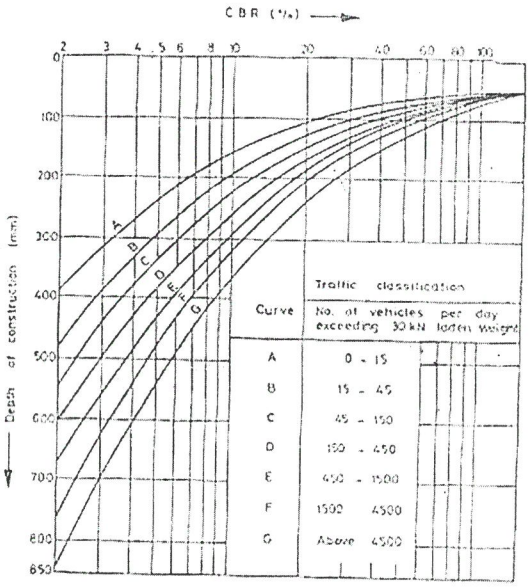

Fig-3.a: CBR design chart recommended by IRC

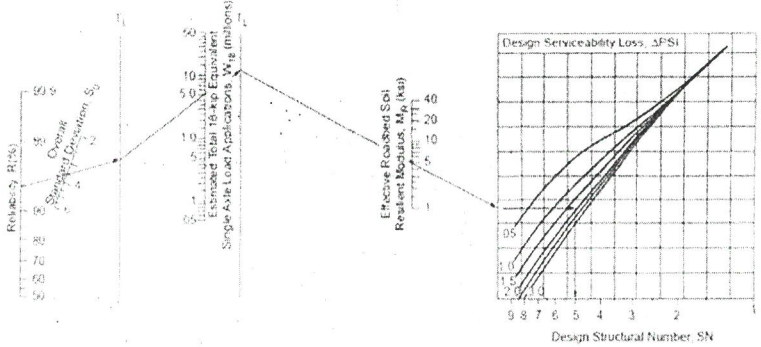

Fig-4.a: Nomograph to solve for SN

Table-4.a: drainage coefficients for untreated granular layers of flexible pavement

Quality of Drainage		% of Time Pavement Structure is Exposed to Moisture Levels Approaching Saturation			
Rating	Water Removed Within	< 1%	1–5%	5–25%	> 25%
Excellent	2 h	1.40–1.35	1.35–1.30	1.30–1.20	1.20
Good	1 day	1.35–1.25	1.25–1.15	1.15–1.00	1.00
Fair	1 week	1.25–1.15	1.15–1.05	1.00–0.80	0.80
Poor	1 month	1.15–1.05	1.05–0.80	0.80–0.60	0.60
Very Poor	Never drain	1.05–0.95	0.95–0.75	0.75–0.40	0.40